

SAR-to-Optical image Translation

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Abstract

Cloud cover remains a persistent challenge in optical remote sensing, limiting the usability of optical satellite imagery for continuous Earth observation. Synthetic Aperture Radar (SAR) data, in contrast, provides cloud-penetrating, all-weather imaging but lacks the spectral and visual richness of optical observations. Bridging these complementary modalities, this study investigates SAR-to-optical image translation using the Pix2Pix conditional generative adversarial network (cGAN). While existing research predominantly focuses on reconstructing only the visible (RGB) or near-infrared bands, this work employs the winter subset of the SEN12-MS dataset to address the full spectral range. The objectives are threefold: (i) to validate SAR-to-optical translation across all 13 Sentinel-2 spectral bands; (ii) to assess the reliability and reconstructability of each individual band; and (iii) to evaluate the performance of the translation model for cloud removal. Experimental results show that the model effectively learns the SAR-to-optical mapping and achieves high reconstruction quality across all spectral bands, though bandwise analysis reveals that reconstruction accuracy varies with spectral characteristics. When applied to the SEN12-MS-CR dataset, the model successfully generates cloud-free optical imagery that closely matches reference data, achieving performance competitive with state-of-the-art models such as DiffCR. Overall, the findings confirm the viability of SAR-to-optical translation for producing spectrally consistent, cloud-free optical imagery, thus enhancing the temporal continuity of Earth observation data. Two ablation studies further analyze the impact of different loss functions and the exclusion of 60 m bands. Additionally, based on data analysis, a new per-band clipping strategy for optical data is proposed.

Keywords: keyword 1; keyword 2; keyword 3 (List three to ten pertinent keywords specific to the article; yet reasonably common within the subject discipline.)

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2. Introduction

Earth observation has become an indispensable tool for understanding and monitoring the planet's dynamic processes. Optical and radar remote sensing represent two complementary modalities at the core of modern monitoring systems. Optical sensors, such as the Multi-Spectral Instrument (MSI) onboard Sentinel-2, provide rich spectral and visual information essential for applications ranging from agricultural monitoring and land-cover classification to disaster response and forest monitoring [1]. However, these sensors are inherently constrained by atmospheric conditions, particularly cloud cover. Global estimates suggest that average cloud cover exceeds 66% [4–6], with about 55% over land surfaces [6], causing considerable data gaps in both spatial and temporal domains. For applications requiring consistent time series, e.g., agricultural monitoring, or where a specific scene must be observed at a given time, e.g., disaster monitoring, cloud cover represents a serious limitation [6]. The diversity of clouds—including thin and thick clouds as well as haze—together with the wide range of occlusion scenarios and their uneven distribution, poses an additional challenge for image reconstruction and the generalizability of cloud removal techniques [7]. In contrast, Synthetic Aperture Radar (SAR) sensors, operating in the microwave domain, such as the C-band instrument on Sentinel-1 mission offer all-weather, day-and-night imaging capabilities independent of solar illumination or cloud interference. However, the backscatter-based nature of SAR imagery introduces challenges related to speckle noise, geometric distortions, and a lack of intuitive spectral color information [2,3], necessitating advanced interpretation skills.

To bridge the gap between these two sensing modalities, SAR-to-optical image translation has emerged as a powerful generative approach. It aims to synthesize optical-like, cloud-free imagery from SAR data, combining the interpretability of optical observations with the reliability of radar acquisitions. The domain gap between SAR and optical imagery, however, poses significant challenges. For example, while two objects with identical structures may appear different in optical imagery due to their spectral responses, they can appear similar in SAR imagery, reflecting SAR's emphasis on structural rather than spectral properties [16]. Moreover, acquiring perfectly co-registered SAR–optical pairs is also non-trivial, as both spatial and temporal alignment must be ensured.

Recent advances in generative artificial intelligence (GenAI), particularly in conditional generative adversarial networks (cGANs) [8] and diffusion models [9], have made it possible to learn complex mappings between SAR and optical domains with visually plausible results [10]. These developments have opened new pathways for applications in land-cover classification, vegetation monitoring, disaster response, and particularly, cloud removal. Despite the rapid progress in this field, a significant gap remains in the literature. Most existing studies have focused primarily on reconstructing the visible RGB subset of optical imagery such as [2,10,11,13,23], with a few extending to the Near-Infrared (NIR) range such as [14,15]. Consequently, the full multispectral potential of missions such as Sentinel-2 remains largely underexplored in the context of generative reconstruction. Critical bands for vegetation analysis (Red-Edge) and soil moisture monitoring (SWIR) are rarely addressed.

Furthermore, while SAR-to-optical translation is often evaluated qualitatively, systematic assessments of its reliability across individual spectral bands remain scarce. Finally, although the approach shows promise for cloud removal, its effectiveness in this context has not yet been comprehensively validated across the full spectral range.

Accordingly, this study investigates the use of generative models for translating Sentinel-1 SAR imagery into the full multispectral stack of Sentinel-2 imagery. The objectives of this work are threefold:

1. To validate SAR-to-optical image translation across all 13 Sentinel-2 spectral bands, moving beyond standard RGB constraints.
2. To systematically assess the reliability and reconstructability of each individual spectral band.
3. To evaluate the performance of the translation model for the practical application of cloud removal.

By addressing these objectives, this work aims to provide a comprehensive and quantitative understanding of SAR-to-optical translation as a multimodal learning problem and to clarify its potential and limitations for enhancing the temporal continuity of Earth observation data.

3. Related Work

3.1. Cloud Removal Methods (Overview)

Cloud removal methods in optical remote sensing are commonly grouped into three categories: (i) single-image methods, (ii) multimodal-based methods, and (iii) multitemporal-based methods [7].

Single-image methods aim to reconstruct surface information using only the cloudy optical image and typically rely on statistical or physical priors (e.g., frequency filtering or atmospheric scattering models). Zhang et al. [33] proposed the Haze Optimized Transformation (HOT) for thin cloud and haze compensation, while He et al. [34] introduced the dark channel prior, later adapted for thin cloud removal in optical remote sensing. With deep learning, CNNs, U-Nets, and GAN-based architectures have also been applied in this setting; however, single-image methods remain fundamentally limited under dense cloud cover due to missing surface information [7].

Multimodal-based methods integrate auxiliary data from complementary sensors to improve restoration. In practice, the most significant progress has been achieved by fusing SAR and optical imagery, leveraging the cloud-penetrating capability of SAR. Meraner et al. [6] proposed DSen2-CR, combining Sentinel-1 and Sentinel-2 to enhance reconstruction under thick clouds and preserve spectral fidelity. Grohnfeldt et al. [5] demonstrated the effectiveness of conditional GANs for SAR–optical fusion, and Xu et al. [35] proposed GLF-CR using a global–local fusion strategy. This line of work motivates SAR-to-optical translation approaches, which either translate SAR features into optical-like imagery or fuse SAR with partially corrupted optical inputs, while still facing challenges related to data registration, modality differences, and SAR speckle.

Multitemporal-based methods exploit repeated observations to fill cloud-covered regions. Xu et al. [28] proposed multitemporal dictionary learning for reconstructing thin and thick cloud regions without explicit cloud masks. Ebel et al. [30] introduced UnCRtainTS, combining multispectral reconstruction with per-pixel uncertainty estimation. While highly effective under persistent cloud cover, multitemporal approaches can be sensitive to geometric misalignment and temporal land-cover changes [36], and they require consistent time series data; mono-temporal approaches avoid such requirements and reduce co-registration constraints [37]. Given the focus of this work on mono-temporal SAR–optical generation, the following subsections review SAR-to-optical translation methods and their use for cloud removal.

3.2. SAR-to-Optical Image Translation

SAR-to-optical translation is defined as an image-to-image (I2I) task that synthesizes optical-like imagery from SAR data, despite the domain gap between backscatter intensity and spectral reflectance as well as SAR-specific artifacts such as speckle and geometric distortions. Before the emergence of generative AI, SAR-to-optical translation relied on heuristic or classical methods. Early approaches used pseudo-colorization of SAR channels or polarization composites to enhance interpretability [17], though they failed to reproduce true optical characteristics. Multisensor fusion methods, such as combining SAR with prior cloud-free optical data via intensity–hue–saturation (IHS) or wavelet-based transforms [18], provided partial solutions but depended on handcrafted features and complex preprocessing, limiting scalability and accuracy.

Recent deep learning approaches—most notably generative adversarial networks (GANs) [19] and diffusion models [9]—have enabled end-to-end learning of mappings from SAR to optical observations by modeling cross-modal feature relationships and pixel-level distributions. Most existing studies focus on reconstructing the visible (RGB) subset of Sentinel-2 (or extend to near-infrared), whereas comparatively fewer works address

full multispectral reconstruction; correspondingly, the literature often distinguishes SAR-to-optical translation from SAR-to-multispectral (SAR-to-MS) translation. In this paper, the term “optical” denotes the full 13-band Sentinel-2 spectral range, and the proposed methodology extends SAR-to-optical translation to the complete multispectral domain.

The foundational GAN framework by Goodfellow et al. [19], later extended to Conditional GANs (cGANs) by Mirza and Osindero [8], enabled image-to-image translation tasks that made SAR-to-optical synthesis feasible. Fuentes Reyes et al. [20] optimized an unsupervised CycleGAN for SAR-to-optical translation, improving interpretability and reducing speckle through customized preprocessing and architectural refinements, although fine structural details remained difficult to preserve in urban scenes. Wang et al. [21] introduced a Supervised CycleGAN (S-CycleGAN) by incorporating a pixel-wise MSE loss, producing realistic optical images and showing strong potential for cloud removal. Gao et al. [10] extended this idea with a fusion-based GAN framework for high-resolution imagery. Instead of directly translating SAR to optical, they generated a simulated optical image from SAR data and fused it with both SAR and optical inputs to reconstruct more realistic results. An ablation study confirmed that this two-stage fusion strategy achieved superior performance among tested configurations.

Following the success of Vision Transformers (ViTs) [22], transformer-based architectures have also emerged. Zhao et al. [23] proposed the Hybrid Vision Transformer cGAN (HVT-cGAN), combining CNN and ViT branches to capture both local details and global semantics. A Convolutional Attention Fusion Module (CAFM) adaptively merged multiscale features, enhancing texture and color fidelity. Trained on the SEN1-2 dataset [24], HVT-cGAN achieved superior visual and quantitative results over previous GAN-based models. Park et al. [25] further improved this approach with a multiscale ViT-based cGAN architecture integrating perceptual loss and a two-phase transfer learning strategy to enhance realism and stability.

Diffusion models have recently entered this field. Bai et al. [11] introduced a conditional diffusion model for SAR-to-optical translation, which, despite limited experimentation due to computational constraints, demonstrated promising performance compared to GAN-based methods. Bai et al. [13] later extended the framework with color supervision to improve reconstruction fidelity. A more recent contribution, the Multi-Temporal Conditional GAN (MTcGAN) by Kwak and Park [26], was designed for early-stage crop monitoring and utilized SAR–optical pairs from reference and prediction dates to capture temporal dynamics, achieving superior spectral consistency compared to conventional methods.

These developments have motivated the use of SAR-to-optical synthesis as a data-driven strategy for cloud removal, discussed next.

3.3. SAR-to-Optical Image Translation for Cloud Removal

SAR-guided cloud removal exploits the cloud-penetrating capability of SAR to reconstruct missing optical observations in cloud-contaminated scenes. Early methods relied on traditional signal processing techniques: Huang et al. [27] introduced sparse representation-based cloud removal using SAR data, which Xu et al. [28] extended via multi-temporal dictionary learning. These approaches demonstrated the feasibility of SAR-assisted cloud reconstruction but struggled under heavy cloud cover or rapidly changing surface conditions.

With deep learning, multimodal fusion approaches became dominant. Enomoto et al. [14] applied a cGAN for cloud removal by fusing the RGB composite of a cloudy optical image with the cloud-free near-infrared (NIR) band to reconstruct a cloud-free RGB image. Although limited under dense cloud conditions, the approach was pivotal in demonstrating

the potential of multimodal data fusion for cloud removal and laid the groundwork for subsequent studies. Building on this concept, Grohnfeldt et al. [5] introduced SAR-Opt-cGAN, a model designed to fuse Sentinel-1 SAR and Sentinel-2 optical data—marking the first use of SAR data within a cGAN framework for cloud mitigation. Their adaptation of the Pix2Pix architecture [29] allowed flexible multi-channel inputs and was trained on a subset of the SEN1-2 dataset [24]. The results confirmed the advantage of incorporating SAR data, validating the effectiveness of the approach for mitigating cloud cover.

Beyond direct cGAN fusion, specialized pipelines were proposed. Darbaghshahi et al. [2] introduced a dual-GAN framework with Dilated Residual Inception Blocks (DRIBs), translating SAR to optical in the first stage and fusing with cloudy optical inputs in the second; it showed strong qualitative results but moderate quantitative accuracy. Ebel et al. [30] advanced this direction with UnCRtainTS, which reconstructs multispectral imagery while predicting per-pixel uncertainty, improving performance and providing reliability indicators.

Recently, diffusion models have become highly competitive. Zou et al. [31] proposed DiffCR, a fast conditional diffusion framework and reported state-of-the-art on SEN12-MS-CR [32]. DiffCR employs a decoupled conditional architecture and a Time and Condition Fusion Block (TCFBlock), and predicts cloud-free images directly, enabling high-quality results in as few as 1–5 denoising steps with substantially reduced computational cost. In parallel, Meraner et al. [6] proposed DSen2-CR, a deep residual SAR–optical fusion network with a Cloud-Adaptive Regularized Loss (CARL) to preserve uncorrupted content while focusing reconstruction on clouded regions; evaluated on SEN12-MS-CR, it showed strong global generalization and high reconstruction accuracy across all 13 bands, with best performance on surface-related bands and higher errors on atmospheric bands.

Overall, the literature demonstrates a progression from classical SAR-guided reconstruction to deep multimodal fusion, uncertainty-aware prediction, and diffusion-based models, establishing SAR-to-optical translation as a strong paradigm for cloud removal. This work builds on that foundation by evaluating SAR-to-optical translation for cloud removal across the full 13-band Sentinel-2 spectrum and analyzing model behavior through bandwise evaluation and ablation studies.

4. Materials and Methods

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Research manuscripts reporting large datasets that are deposited in a publicly available database should specify where the data have been deposited and provide the relevant accession numbers. If the accession numbers have not yet been obtained at the time of submission, please state that they will be provided during review. They must be provided prior to publication.

Interventionary studies involving animals or humans, and other studies require ethical approval must list the authority that provided approval and the corresponding ethical approval code.

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5.1. Subsection

5.1.1. Subsubsection

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1. First item;
2. Second item;
3. Third item.

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¹ Tables may have a footer.

Table 2. Sentinel-2 MSI spectral bands with central wavelength and spatial resolution

Band	Central Wavelength [nm]	Resolution [m]
B1	443 (Aerosols)	60
B2	490 (Blue)	10
B3	560 (Green)	10
B4	665 (Red)	10
B5	705 (Red edge)	20
B6	740 (Red edge)	20
B7	783 (Red edge)	20
B8	842 (NIR)	10
B8a	865 (Red edge)	20
B9	945 (Water vapour)	60
B10	1375 (Cirrus)	60
B11	1610 (SWIR)	20
B12	2190 (SWIR)	20

The text continues here (Figure 2 and Table 3).

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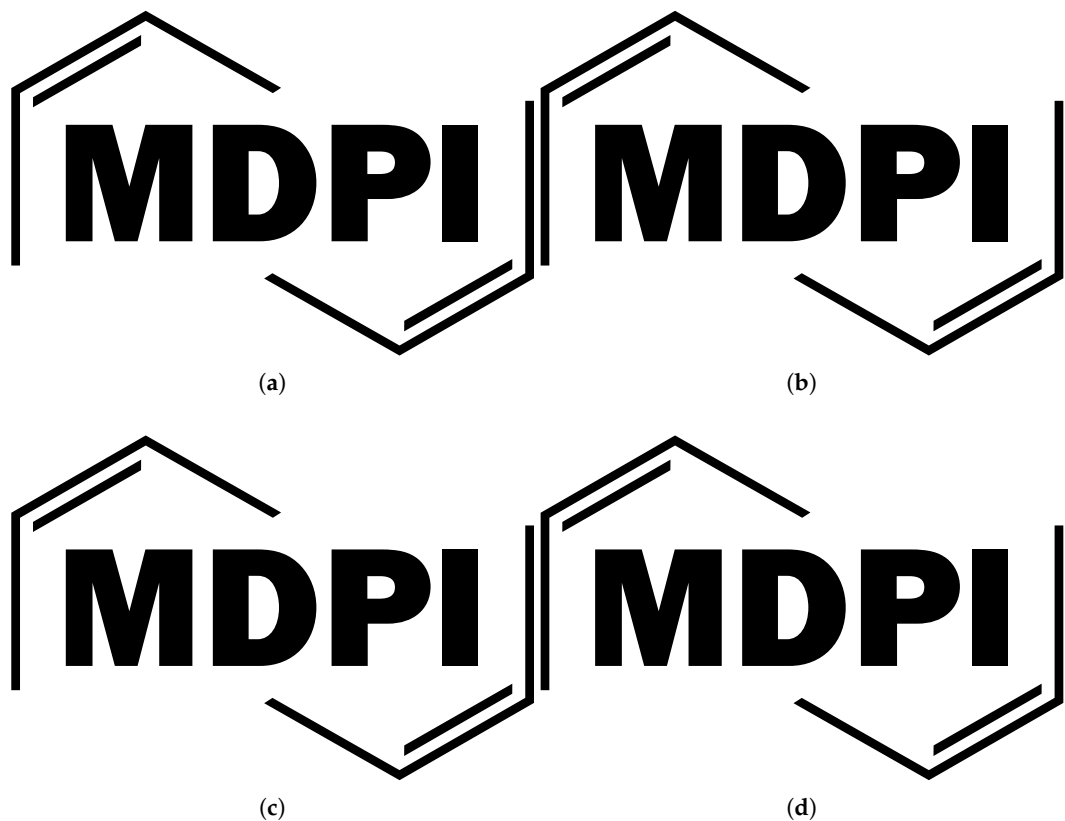


Figure 2. This is a wide figure. Schemes follow the same formatting. If there are multiple panels, they should be listed as: (a) Description of what is contained in the first panel. (b) Description of what is contained in the second panel. (c) Description of what is contained in the third panel. (d) Description of what is contained in the fourth panel. Figures should be placed in the main text near to the first time they are cited. A caption on a single line should be centered.

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Entry 2	Data	Data	Data
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* Tables may have a footer.

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5.3. Formatting of Mathematical Components

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This is the example 1 of equation:

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$$a = 1,$$

(1)

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the text following an equation need not be a new paragraph. Please punctuate equations as regular text.

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This is the example 2 of equation:

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$$a = b + c + d + e + f + g + h + i + j + k + l + m + n + o + p + q + r + s + t + u + v + w + x + y + z$$

(2)

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Please punctuate equations as regular text. Theorem-type environments (including propositions, lemmas, corollaries etc.) can be formatted as follows:

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Theorem 1. *Example text of a theorem.*

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Proof of Theorem 1. Text of the proof. Note that the phrase “of Theorem 1” is optional if it is clear which theorem is being referred to. □

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Abbreviations

The following abbreviations are used in this manuscript:

- MDPI Multidisciplinary Digital Publishing Institute
- DOAJ Directory of open access journals
- TLA Three letter acronym
- LD Linear dichroism

Appendix A

Appendix A.1

The appendix is an optional section that can contain details and data supplemental to the main text—for example, explanations of experimental details that would disrupt the flow of the main text but nonetheless remain crucial to understanding and reproducing the research shown; figures of replicates for experiments of which representative data are shown in the main text can be added here if brief, or as Supplementary Data. Mathematical proofs of results not central to the paper can be added as an appendix.

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Appendix B

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